GOLD PRICE PREDICTION

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DATASET OVERVIEW

- Historical Gold Futures data with daily frequency.
- Contains key price indicators: Open, High, Low, Close, Adj Close, Volume.
- Close price used as the primary target for forecasting.
- Time-series dataset with multi-year span, ideal for sequence modeling.
- Cleaned and sorted chronologically to maintain temporal integrity.

Date	Open	High	Low	Close	Adj Close	Volume
2020-11-16 05:00:00	1874.5999755859400	1887.300048828130	1871.0	1887.300048828130	1887.300048828130	6.0
2020-11-17 05:00:00	1888.4000244140600	1888.4000244140600	1884.5	1884.5	1884.5	59.0
2020-11-18 05:00:00	1873.5	1873.5	1873.5	1873.5	1873.5	152.0
2020-11-19 05:00:00	1865.800048828130	1865.800048828130	1856.0	1861.0999755859400	1861.0999755859400	59.0
2020-11-20 05:00:00	1871.199951171880	1872.5999755859400	1871.199951171880	1872.5999755859400	1872.5999755859400	1900.0
2020-11-23 05:00:00	1834.800048828130	1837.800048828130	1829.199951171880	1837.800048828130	1837.800048828130	843.0
2020-11-24 05:00:00	1829.300048828130	1829.300048828130	1799.300048828130	1804.800048828130	1804.800048828130	1041.0
2020-11-25 05:00:00	1801.4000244140600	1805.699951171880	1801.4000244140600	1805.699951171880	1805.699951171880	205754.0

Layer (type)	Output Shape	Param #
simple_rnn_1 (SimpleRNN)	(None, 60, 128)	16,640
dropout (Dropout)	(None, 60, 128)	0
simple_rnn_2 (SimpleRNN)	(None, 64)	12,352
dropout_1 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 1)	65

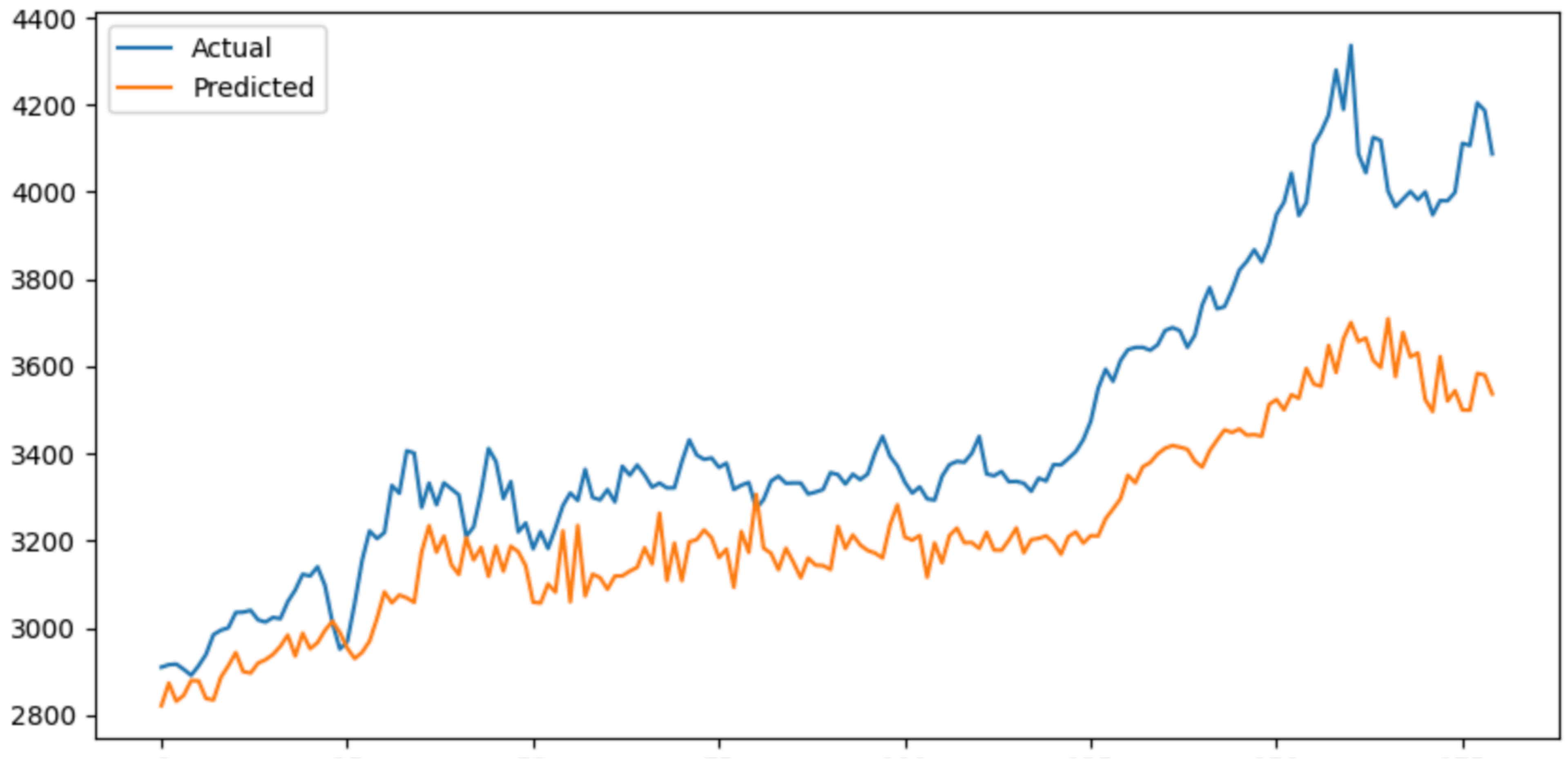
Total params: 29,057 (113.50 KB)
Trainable params: 29,057 (113.50 KB)
Non-trainable params: 0 (0.00 B)

RNN MODEL

- Built using stacked SimpleRNN layers with hidden sizes of 128 → 64, enabling learning of sequential price patterns.
- Dropout layers added after each RNN layer to reduce overfitting and improve generalization.
- Final Dense layer (1 unit) outputs the next-day predicted closing price.
- Contains ~29K trainable parameters, making it lightweight but limited in long-term memory.
- Serves as a baseline sequence model, showing how advanced models (LSTM/GRU) improve over standard RNNs.

RNN MODEL INSIGHTS

- Acts as the baseline model for sequence forecasting.
- Learns short-term patterns but struggles with long-term dependencies.
- Suffers from vanishing gradients, leading to weaker performance on volatile financial data.
- Produces smoother, less accurate predictions compared to LSTM and GRU.
- Useful mainly for comparison to show the advantage of advanced architectures.




```
RNN RMSE: 272.069384532696
RNN MAE: 229.83074951171875
RNN MAPE: 6.356969
```

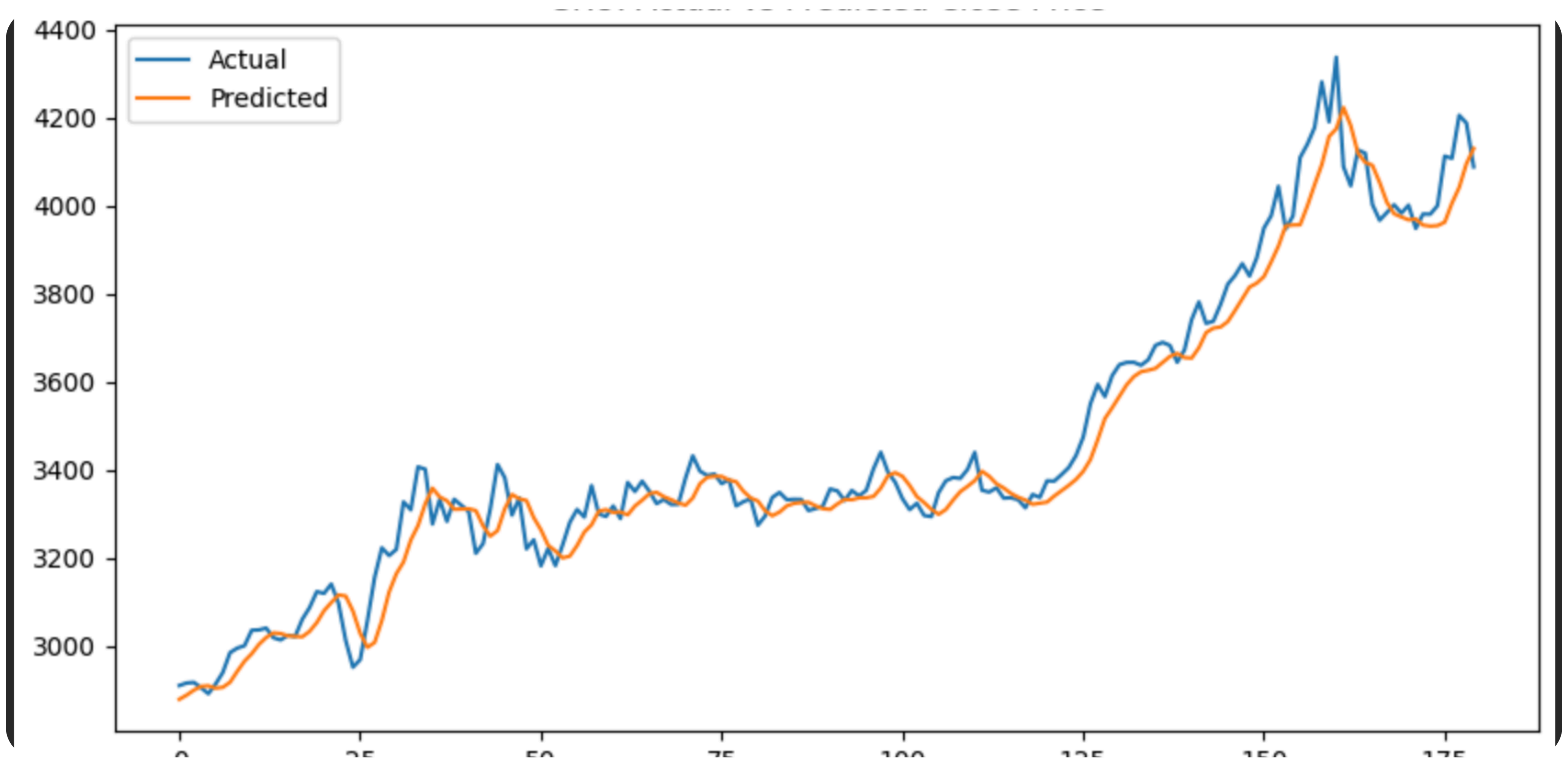
Layer (type)	Output Shape	Param #
gru_8 (GRU)	(None, 60, 128)	50,304
dropout_8 (Dropout)	(None, 60, 128)	0
gru_9 (GRU)	(None, 64)	37,248
dropout_9 (Dropout)	(None, 64)	0
dense_4 (Dense)	(None, 1)	65

```
Total params: 87,617 (342.25 KB)
Trainable params: 87,617 (342.25 KB)
Non-trainable params: 0 (0.00 B)
```

GRU MODEL

- Two-layer GRU architecture with 128 and 64 units, enabling the model to capture both short-term and long-term temporal dependencies in the sequence data.
- Dropout regularization applied after each GRU layer to prevent overfitting and improve the model's generalization on unseen data.
- Dense output layer with a single neuron for final prediction, ideal for regression-based time-series forecasting.
- Total parameters: 87,617, making the model computationally efficient while maintaining strong learning capacity.
- EarlyStopping + ReduceLROnPlateau callbacks used to optimize training, avoid overfitting, and ensure the model converges smoothly.

GRU MODEL INSIGHTS



- Achieved low error values (MAPE ~1.4%), showing strong prediction accuracy.
- Predicted curve closely follows the actual trend, indicating excellent fit and generalization.
- GRU effectively captures both short-term variations and long-term patterns in the data.
- Suitable for time-series forecasting due to stable and smooth predictions.

```
GRU RMSE : 64.80756143793138
GRU MAE  : 49.33497619628906
GRU MAPE : 1.4026474
```

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 60, 128)	66,560
dropout (Dropout)	(None, 60, 128)	0
lstm_1 (LSTM)	(None, 64)	49,408
dropout_1 (Dropout)	(None, 64)	0
dense (Dense)	(None, 1)	65

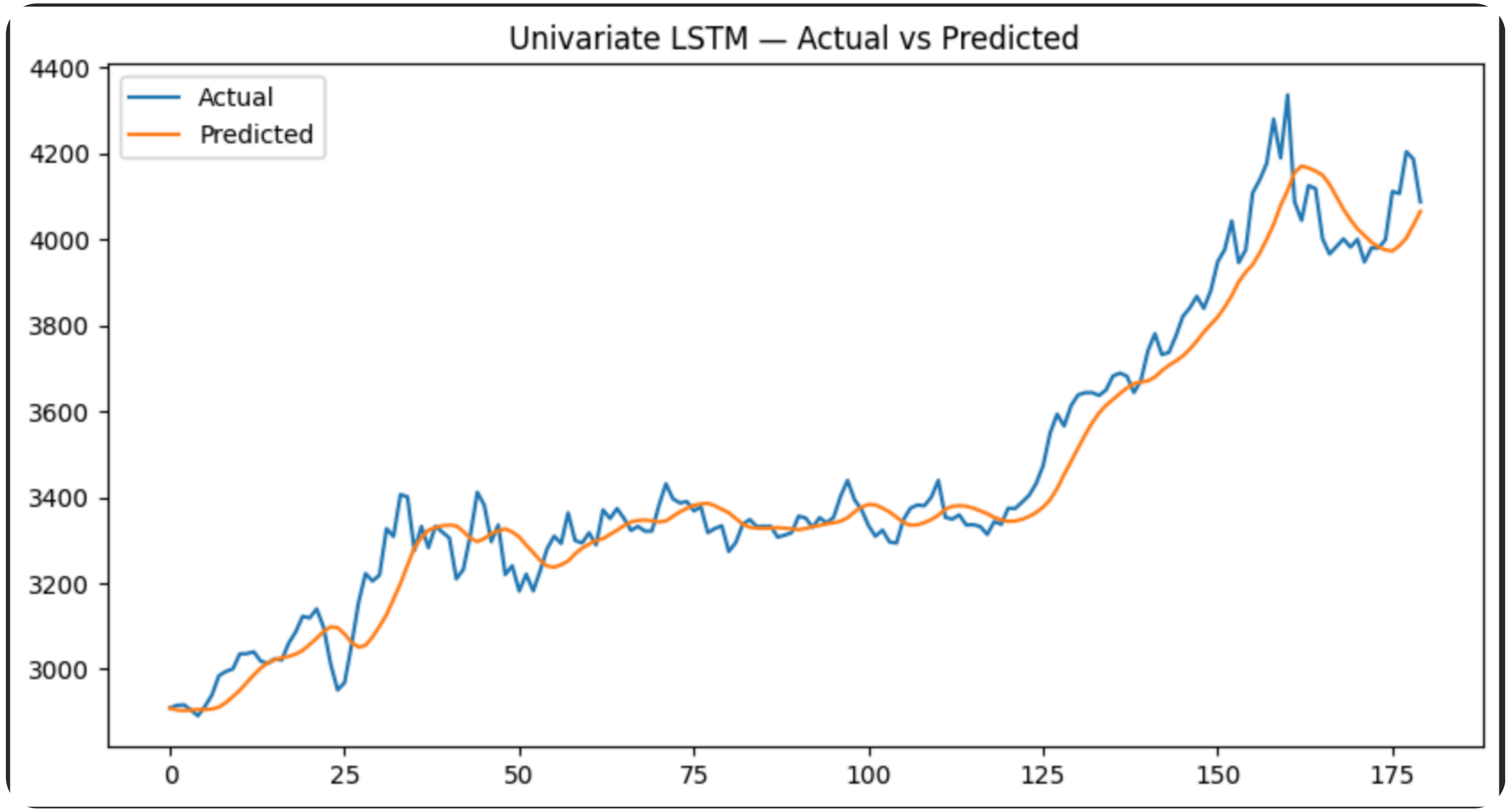
Total params: 116,033 (453.25 KB)
Trainable params: 116,033 (453.25 KB)
Non-trainable params: 0 (0.00 B)

LSTM MODEL

- Built using a two-layer LSTM architecture (128 → 64 units) to capture long-term dependencies in the time-series data.
- Dropout layers added after each LSTM layer to reduce overfitting and improve model generalization.
- Contains ~116K trainable parameters, offering higher learning capacity compared to GRU.
- Designed for sequence forecasting tasks, making it effective for capturing complex temporal patterns.

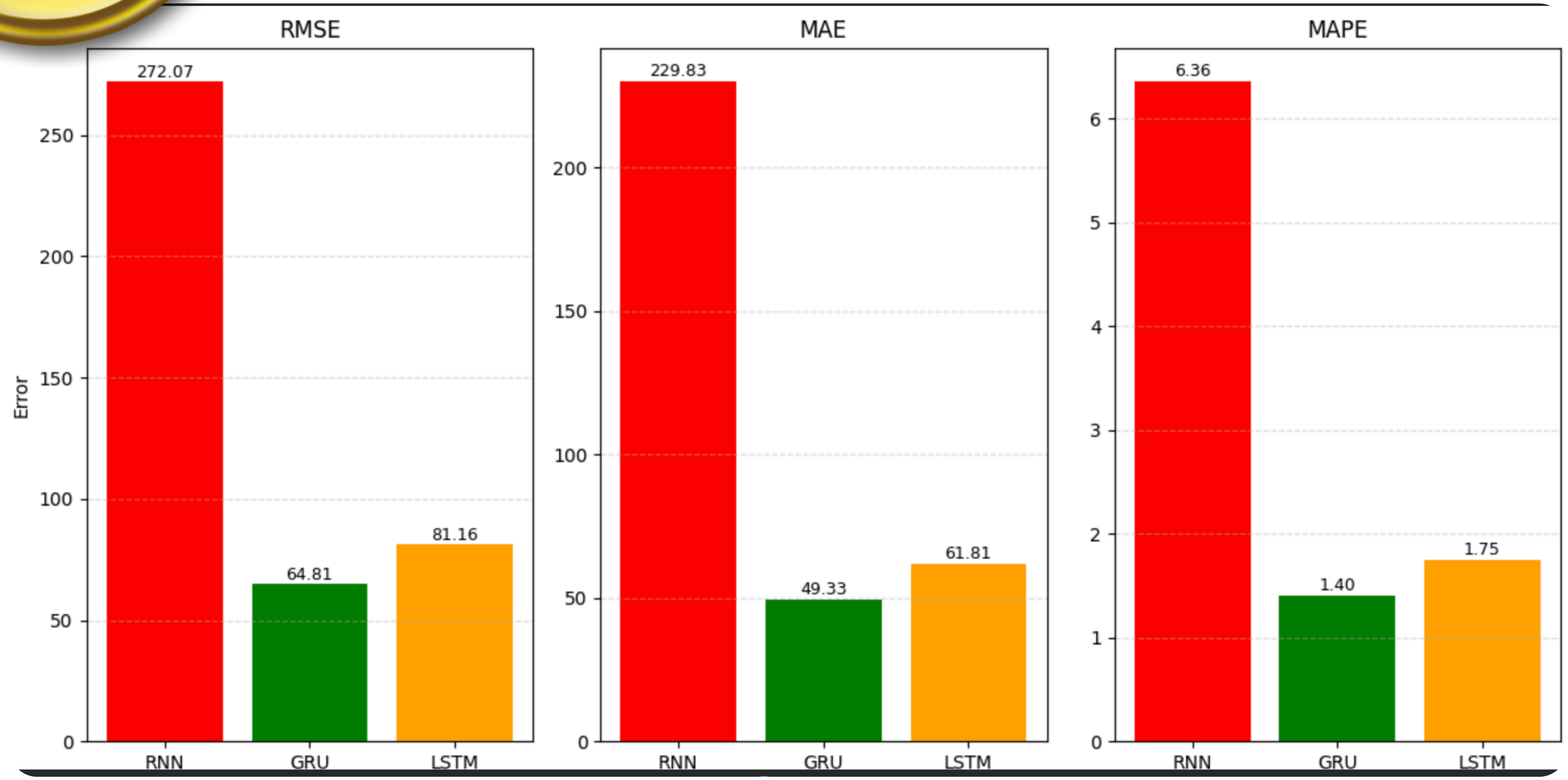
LSTM MODEL INSIGHTS

```
RMSE: 81.155940146693
MAE: 61.810287475585
MAPE: 1.7481202
```



- **LSTM closely follows the overall trend of the actual prices.**
- **Model predictions are smoother, reducing noise from the data.**
- **Achieved low errors: RMSE ~81, MAE ~61, MAPE ~1.74%.**
- **Performs well in stable and upward-trending periods.**
- **Slightly struggles during sharp, volatile price spikes.**

MODEL COMPARISONS

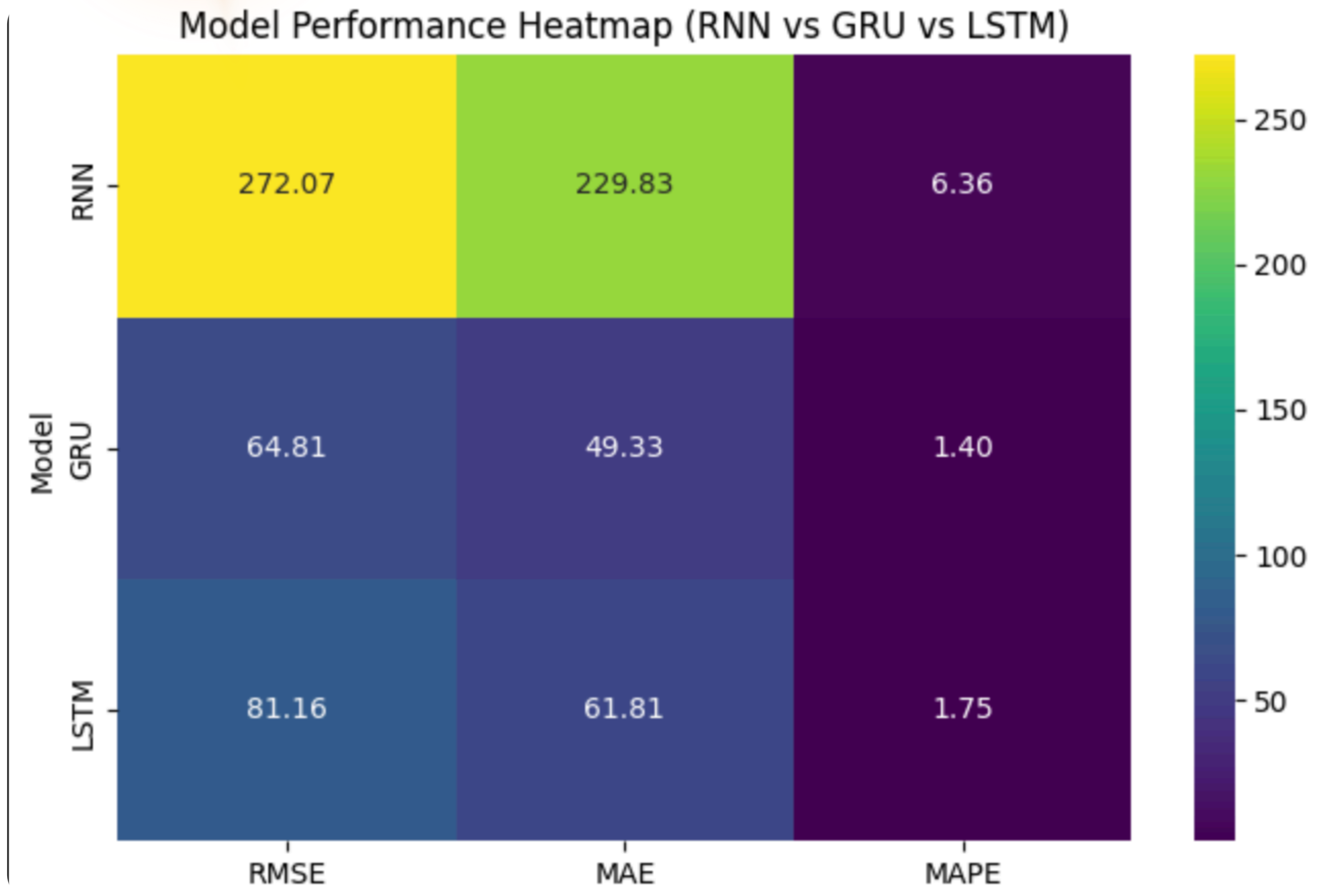


Why GRU performs better than LSTM

- GRU is simpler and less computationally heavy, making it easier to train on medium sized datasets.
- Fewer gates → faster learning, which helps when patterns are not extremely complex.
- The dataset's trend patterns are not highly chaotic, so GRU's simpler architecture models them more efficiently than LSTM.

- RNN shows the highest errors across all metrics → struggles to learn long-term patterns.
- LSTM performs much better than RNN → captures trends well but slightly smooths predictions.
- GRU achieves the lowest RMSE, MAE, and MAPE → best overall performance on this dataset.

INSIGHTS



- Deep learning models (LSTM/GRU) provide more reliable gold price forecasts, helping investors make data-driven buying and selling decisions.
- Improved trend and volatility detection enables better risk management for traders, brokers, and commodity analysts.
- GRU's faster training and strong accuracy make it suitable for real-time forecasting systems used in trading platforms.
- Forecasting supports strategic planning for industries dependent on gold (jewelry, electronics, banking) by predicting cost fluctuations.
- RNN-based models highlight future price direction, helping businesses optimize hedging strategies and reduce financial exposure.

THANK YOU